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SERIES: GRADING OF RECOMMENDATIONS ASSESSMENT, DEVELOPMENT AND EVALUATION (GRADE)

GRADE Guidelines: 16. GRADE evidence to decision frameworks for tests in clinical practice and public health

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Abstract

Objectives: To describe the grading of recommendations assessment, development and evaluation (GRADE) interactive evidence to decision (EtD) frameworks for tests and test strategies for clinical, public health, or coverage decisions.

Study Design and Setting: As part of the GRADE Working Group's DECIDE project, we conducted workshops, user testing with systematic review authors, guideline developers and other decision makers, and piloted versions of the EtD framework.

Results: EtD frameworks for tests share the structure, explicitness, and transparency of other EtD frameworks. They require specifying the purpose of the test, linked or related management, and the key outcomes of concern for different test results and subsequent management. The EtD criteria address test accuracy and assessments of the certainty of the additional evidence necessary for decision making. When there is no direct evidence of test effects on patient-important outcomes, formal or informal modeling is needed to estimate effects. We describe the EtD criteria based on examples developed with GRADEpro (www.gradepro.org), GRADE's software that also allows development and dissemination of interactive summary of findings tables.

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Conclusion: EtD frameworks for developing recommendations and making decisions about tests lay out the sequential steps in reviewing and assessing the different types of evidence that need to be linked. © 2016 Elsevier Inc. All rights reserved.

Keywords: Guidelines; Recommendations; Decision-making; Tests; GRADE; GRADEpro; Summary of findings

1. Background

An essential component of formulating recommendations or making decisions about the best care for patients and populations is the appropriate use of tests. Tests have many purposes including establishing a diagnosis, screening of a population, and prognosticating or monitoring a disease or condition. Different types of tests, for example, radiological, laboratory, or psychological evaluations, are available for these purposes. Regardless of the purpose, tests can fulfill different roles; they can supersede an existing reference test (or sequence of tests) or be compared against an existing commonly used test (or sequence of tests) as a triage, replacement, add-on, or parallel tests [1,2]. For simplicity and to account for the differing purposes, here we will use the term "test" regardless of the type, role, or number of tests used.

In this article, we will describe the special considerations that are required when making decisions and recommendations about tests and test strategies [3]. The grading of recommendations assessment, development and evaluation (GRADE) evidence to decision (EtD) criteria for clinical recommendations, public health, or coverage decisions about health interventions apply to tests [3,4]. However, tests by themselves usually affect outcomes indirectly rather than directly [4]. Tests should not be applied without carefully considering the intended and unintended consequences, including the presumptive management strategies and outcomes that follow from applying a test [5–7]. It is primarily downstream management—guided by the results of the test or a series of tests—that affects patient-important outcomes. Consequently, decisions about using a test for a specific purpose require identification of the relation between test results, downstream actions, and outcomes.

1.1. Why are EtD frameworks needed for tests?

Boxes 1 and 2 present a scenario illustrating the link between testing and subsequent management. Ideally, decisions, such as those about health recommendations, that involve tests should be informed by evidence that directly addresses all the important effects of both testing and subsequent management strategies. When such direct evidence is available, EtD frameworks for clinical or public health recommendations or for coverage decisions can be used [4]. Unfortunately, evidence from randomized trials or well-done nonrandomized studies that directly evaluate the impact of tests on important outcomes rarely exists [8]. Thus, decisions about using tests typically require linking evidence about test accuracy with evidence of outcomes of subsequent actions.

Evidence of the effects of tests often comes from test accuracy studies, that is, studies evaluating if a test correctly or incorrectly classifies tested people. Estimating the effects of tests on outcomes based on accuracy studies requires additional assumptions and modeling, to express the benefits and harms of those classified correctly and incorrectly. For example, patients who have a condition but test negative (false negatives) will experience the complications or natural course of a disease that remains undetected, even if only temporarily. In the example in Box 1, patients are at risk of complications from a deep venous thrombosis (DVT) that may not be detected. Estimates of outcomes for these patients can come from prognostic studies or from the control arm of treatment trials. These patients also may undergo further unnecessary testing for other, inappropriately suspected conditions that present with features similar to the condition of interest. Patients who have a condition and test positive (true positives), on the other hand, will experience the benefits and harms of whatever treatment they receive. Estimates of outcomes for these patients come from randomized trials or other evaluations of the effects of those treatments. Evidence about the treatment effects is linked to the estimated number of true positive test results (Box 2 and Fig. 1).

Decision analytical approaches are needed to put together these different types of evidence, to calculate or model the consequences of a decision about a test and assess the certainty of the evidence. EtD frameworks for tests, therefore, include additional criteria to assess the accuracy of tests and the certainty of the additional types of evidence used to estimate the effects of tests on outcomes that are important to people. These additional criteria become relevant for other EtD frameworks (e.g., coverage decisions) when making decisions about tests. The goal of this article is to describe the special considerations that are required when panels make recommendations and decisions about tests in the GRADE EtD frameworks. Understanding the criteria that are used to make these decisions can help develop other tools for decision making in other contexts, for example, by patients and providers as decision aids. However, this article focuses on the use of EtD frameworks by panels or groups of people.

2. Methods

The development of the EtD frameworks started with the GRADE approach to making judgments about the strength

What is new?

Key findings

- The GRADE working group has expanded the criteria for their evidence to decision frameworks. The general criteria that apply to decisions about clinical recommendations and coverage decisions also apply to tests. However, tests by themselves usually affect outcomes indirectly rather than directly. Downstream clinical management, which is guided by the results of the test, affects outcomes. Consequently, decisions about diagnostic tests for a specific purpose require identification of the link between test results and downstream clinical actions.

What this adds to what was known?

- The GRADE working group has previously established criteria for assessing the certainty in evidence about test accuracy studies. This article describes criteria for using test accuracy studies for guideline developers and other decision makers.

What is the implications and what should change now?

- Developing an appropriate model for making a decision about a diagnostic intervention requires specifying the purpose of the test and the key outcomes of concern for different test results.
- In the absence of direct evidence from a randomized trial, performance of a test must be considered. Sometimes these considerations are sufficient for making a decision. GRADE evidence to decision frameworks enhance transparency in the decision-making context, the GRADEpro database of evidence tables and evidence to decision frameworks allow sharing of information for trustworthy decision making in health care.

of recommendations and evidence-to-recommendation tables for health interventions [3,4,9–15] and specifically for tests [16–20]. We further developed the EtD frameworks for diagnostic and other tests using an iterative approach as outlined elsewhere [3] and described in the DECIDE project protocol [21]. This included brainstorming and discussions by the authors and other members of the DECIDE project and the GRADE Working Group, workshops with guideline developers and other decision makers, user testing with systematic review authors and guideline developers, and piloting versions of the framework in guideline development projects. We used the EtDs in over 100 recommendations with various panels and modified it based on their use with these panels [22–24]. Examples of required modifications to the EtD for tests included that the development of questions requires the use of analytical population, intervention, comparator, and outcomes (PICO) frameworks describing pathways to facilitate decision analysis as one part of the EtD framework for diagnostic and other tests [25]. Furthermore, it became apparent that a separate criterion for test accuracy and linked benefits and harms that result from patient management should be inserted. This also required describing the different certainty of evidence for test accuracy and linked evidence. The presentation of complete evidence profiles about test accuracy and management effects within the EtD became also unwieldy and challenging for users to understand. We therefore made changes to the original framework and retested it in workshops and guidelines. A glossary of terminology used in EtD frameworks is available in Appendix 1. To demonstrate the current version of EtD frameworks for tests, we selected two examples for EtD frameworks that address a clinical and a public health question, respectively, and created interactive summary of findings tables and EtD frameworks in GRADEpro (www.gradepro.org) in Box 2, Appendices 2 and 3. The emphasis in this article is on the EtD framework and their use but not on the health care issues addressed. Judgments may appropriately differ, but the EtD framework makes these judgments transparent to users including those who disagree.

3. Results

EtD frameworks for tests have the same structure as EtD frameworks for other recommendations and decisions [3]. This includes formulation of the question, an assessment, and conclusions (see Appendix 1 for a glossary). Questions may be formulated by guideline developers or others making a decision about tests (a guideline panel or a policy maker). The assessment is completed by those reviewing the evidence (e.g., a team of systematic review authors or those assessing a technology) and decision makers (e.g., those making judgments). Conclusions are drawn by those making a recommendation or decision. The recommendation or decision is intended to be implemented by consumers of the EtD.

3.1. Formulating the question

Decisions about tests and management strategies should be informed by the best available evidence of the effects of those decisions on important outcomes. Decisions based on separate evidence about test accuracy and about downstream consequences require developing an appropriate model, specifying the features of a test and outcomes of concern for different test results and subsequent management. Thus, formulating a question about a test or test strategy requires clarifying the problem in a background, the purpose, type

Box 1 Clinical scenario: D-dimer and compression ultrasound for suspected deep venous thrombosis (DVT) in patients with a low pretest probability of first lower extremity DVT [37].

Deep vein thrombosis (DVT) is a common condition that affects approximately one in 1,000 persons per year. Testing for DVT is crucial because clinical assessment alone is unreliable and the consequences of misdiagnosis are serious, including fatal pulmonary embolism.

Three categories of tests are typically used to determine the probability of DVT:

1. Clinical probability assessment based on patient history and clinical findings
2. D-dimer assays
3. Imaging studies (most commonly venous ultrasonography and less frequently venography, CT scan, or MRI)

Diagnostic testing often requires combining the results of more than one assessment. The goal of choosing one strategy over another is to improve patient outcomes in the most efficient manner. In this scenario, we will focus on two issues with an emphasis on the second: (1) whether or not a moderately or highly sensitive D-dimer test should be used to triage patients with suspected DVT but a low clinical probability, so that only those with a positive D-dimer test would undergo further testing; and (2) after a positive D-dimer test, should further testing be done with either compression ultrasound, whole leg ultrasound or venography.

D-dimer, a degradation product of cross-linked fibrin, is typically elevated in patients with acute DVT. However, because D-dimer levels may also be increased in a variety of nonthrombotic disorders (e.g., malignancy, disseminated intravascular coagulation, increasing age, infection, pregnancy, following surgery or trauma, inflammatory conditions, atrial fibrillation, and stroke), D-dimer is a sensitive but nonspecific marker for venous thromboembolism.

MRI, magnetic resonance imaging; CT, computed tomography.

and role of the test, the linked alternative intervention(s), the main outcomes, the setting, and the perspective (Box 2).

Formulating a question about a test in the PICO format [26] should include the test and subsequent management strategies as pathways to important outcomes as well as identifying subgroups that might require different recommendations or options [25]. Figure 1 and Box 2 show the PICO analytical pathways for the scenario as they may be developed by a guideline panel. The use of the acronym PICO serves as a reminder to specify patient-important outcomes, rather than relying on test accuracy as a surrogate outcome, and that the use of tests is an intervention with potentially important consequences [19,27].

In the above scenario, both the intervention and the comparison include diagnostic tests and treatment based on their results. Patients with a suspected DVT but low probability of a DVT will undergo testing with a moderately to highly sensitive D-dimer blood test. If positive, the test would be followed by further testing with compression ultrasound (CUS) of the proximal leg, to confirm or refute the findings, because false positives are common. Alternatives to CUS include the use of venography—the reference test that is more invasive and costly or an ultrasound of the whole leg that also is more costly and, therefore, typically avoided. In those with a positive CUS test result, the probability of disease is now high enough to begin treatment. Those with a negative D-dimer or a positive D-dimer and negative subsequent CUS test would receive watchful waiting during which no treatment is administered. The main patient-important outcomes are death, pulmonary embolism (PE), and intracranial or other major bleeding—not the accuracy of the tests. We will describe previously how the frequency of the outcomes across the test and treatment strategies is influenced by the number of true and false positives and the true and false negatives.

3.2. Making an assessment

Appendix 2 is an EtD framework that summarizes decision makers' assessment for the scenario in Box 1. It illustrates the criteria used for making recommendations about using tests from an individual perspective [3]. The evidence should be summarized by a group responsible for the systematic review or Health Technology Assessments (HTA) with input from the decision maker (e.g., a guideline panel). The decision makers are responsible for the judgments that are made. In the examples provided here, the judgments are hypothetical judgments of a guideline panel.

3.2.1. The problem

As with other recommendations, decision makers need to determine if the problem is a priority. This will depend on the perspective taken (e.g., that of an individual patient or a population [e.g., health system]) and be influenced by the condition for which the test is used and the tests being compared. In the context of new tests, considerations include the limitations of currently used tests in the context of a health care problem that needs to be addressed, such as their accuracy, adverse effects, availability, or costs.

For questions about screening, the number of people that would have to be screened and the number of people who could potentially benefit from screening is also relevant from a public health perspective. The more people at risk of the problem who could potentially benefit from screening, the more likely it is that addressing the problem would be a priority, if the question is screening versus no screening. For example, cervical cancer is a fairly common and treatable problem. If there is low utilization of screening due to, for example, use of an invasive test, improving utilization might be a priority, which could mean assessing the balance of desirable and undesirable

Box 2 Evidence to decision (EtD) framework—Question section[a]

Should D-dimer and compression ultrasound be used for suspected deep venous thrombosis (DVT) in patients with a low pretest probability of first lower extremity DVT? For an online version of the related interactive Summary of Findings Table and EtD framework see the following link to the GRADE database of evidence summaries: http://dbep.gradepro.org/profile/E6E2B650-7F68-405D-96C6-8D2CE131CD9D

Question details	
Patients	Patients with suspected DVT with a low pretest probability
Intervention	D-dimer testing followed by compression ultrasound
Comparison	D-dimer testing followed by whole leg ultrasound or venography
1. Purpose of the tests	Diagnosis of DVT
2. Role of the tests	Triage (D-dimer) and replacement (compression ultrasound compared with whole leg ultrasound or venography)
3. Linked treatment	Anticoagulation with low molecular heparin or warfarin in patients with DVT confirmed by compression ultrasound or watchful waiting in patients with (a) negative D-dimer or (b) positive D-dimer followed by negative compression ultrasound
Main outcomes	Death from pulmonary embolism, nonfatal pulmonary embolism, nonfatal intracranial and other major bleeding, leg pain (from DVT), leg swelling (from DVT), allergic reactions and nephrotoxicity from contrast dye (those undergoing venogram), burden of treatment
Setting	Emergency department outpatient setting
Perspective	Clinical recommendation from population perspective
Subgroups	None

Background

Deep venous thrombosis (DVT) of the lower extremity is a common problem with increased prevalence in patients with risk factors such as cancer or immobilization after surgery. Complications of DVT include death, pulmonary embolism, pain, and local reactions. One of the diagnostic approaches to DVT in patients with low pretest probability, for example, those with mild symptoms but no other risk factors, is the use of D-dimer followed by compression ultrasound (CUS) of the lower leg. Those with positive D-dimer will usually undergo CUS and receive treatment with low molecular heparin or warfarin if the CUS is suspicious for DVT. If D-dimer is negative or the consequential CUS is negative in those with positive D-dimer no further test or treatment may be administered. Treatment, although effective in preventing extension of DVT or pulmonary embolism, comes with a risk of bleeding and other adverse outcomes. Patients who are falsely identified as having a DVT when they do not (false positives) undergo unnecessary treatment and worry unnecessarily. Patients who are falsely identified as not having a DVT (false negatives) will possibly suffer the complications of DVT. A guideline panel addressed the question "In patients with a low pretest probability of first lower extremity DVT, i.e. those without identifiable risk factors, what is the impact of using D-dimer and CUS compared to D-dimer testing followed by whole leg ultrasound or venography for the diagnosis of DVT?"

[a] Templates used for EtD frameworks are adapted for specific types of decisions. The one shown here is for a test from a population perspective.

consequences of a less invasive test strategy. Appendix 3 is an example of an EtD framework from a population perspective for a question about screening for which there is no direct evidence of the effects of the choice of the screening test on downstream consequences.

3.2.2. Test accuracy

Judgments about test accuracy should be based on a summary of findings from a systematic review of test accuracy studies. When a new test is compared against an existing test or test strategy, a very inaccurate test is unlikely to require detailed consideration of other criteria in the EtD framework because its low overall accuracy may be sufficient for making a decision against its use in this context. For example, the accuracy of a serologic test for pulmonary tuberculosis compared with conventional sputum culture is associated with low sensitivity (59%–76%) and a specificity of 91% to 95%. Considering the proportion of false negatives and other possible consequences, this low accuracy may be sufficient not to use this test to diagnose tuberculosis. In most situations, however, one should look for a detailed evaluation of other criteria, including modeling of the impact of test accuracy on patient-important outcomes.

3.2.3. Benefits and harms

Judgments about the benefits and harms of using a test require preparation of a summary of findings for the modeled desirable and undesirable effects on health outcomes. Evidence of the estimates that are used in the model should, so far as possible, come from up-to-date systematic reviews. This includes information about direct benefits and harms of the test (e.g., faster diagnosis, direct adverse effects, or burden of the test) and the downstream effects of interventions. In particular, judgments about the effects of

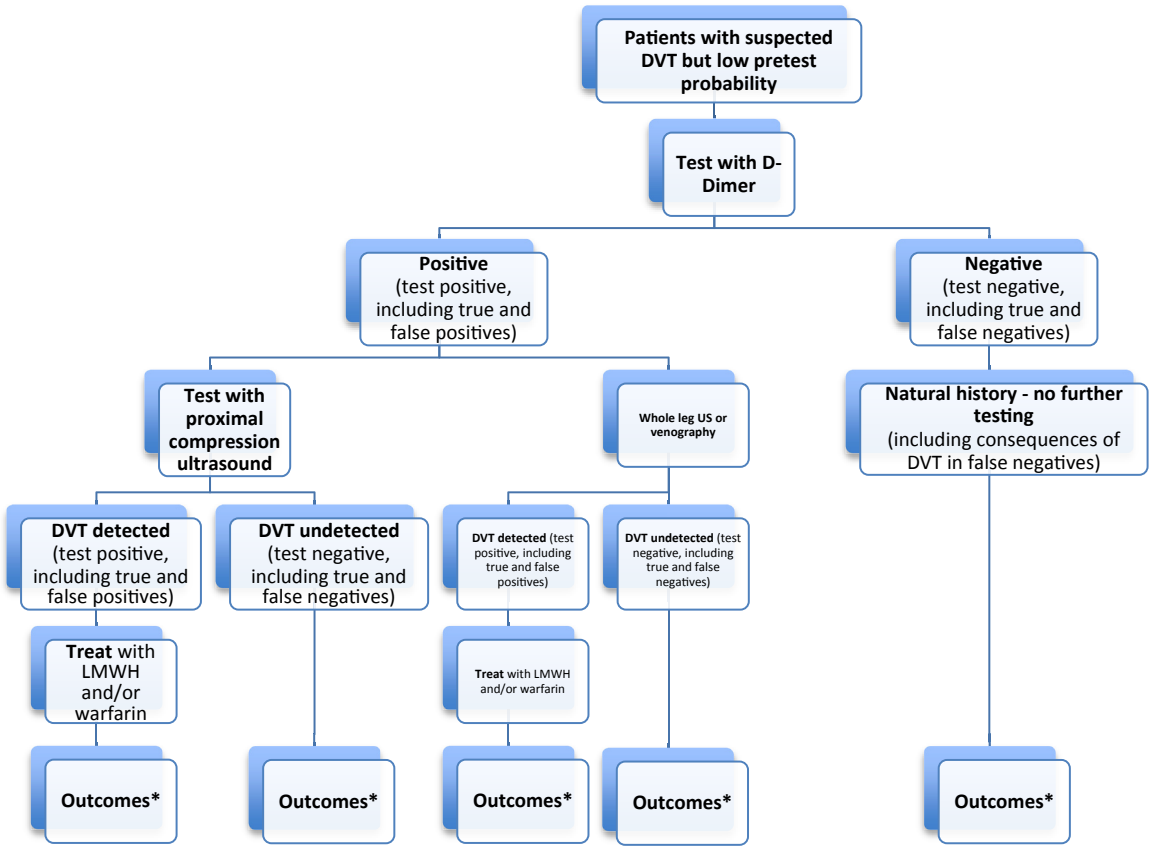

Fig. 1. Analytical PICO framework and pathways to test and treat for suspected deep venous thrombosis (low pretest probability). PICO, population, intervention, comparator, and outcomes.

the interventions that follow based on the test results (linked evidence) should be informed by a summary of findings table [28]. The ideal approach is to have a fully developed model, with detailed assumptions and calculations for transparency (Appendix 3).

When detailed models or calculations are not available, decision makers must make assumptions based on the perceived benefits and harms, providing additional considerations that inform their judgments. For example, a WHO guideline panel considered the consequences of applying serologic tests in a population with a 10% risk of pulmonary tuberculosis where a sensitivity of 59% and specificity of 95% would lead to 81 per 1,000 false positives and 36 per 1,000 false negatives [29]. To evaluate the possible benefits and harms, the guideline panel referred to evidence synthesized in tuberculosis treatment guidelines. It also used a case scenario to describe the potential implications for a country, for example, the resource implications if the test was used as per current practice. This case scenario and the evidence informed the panel's judgments about the consequences of different test results. These consequences included treatment effects, possible detrimental effects from delayed diagnosis, confusing other respiratory diseases with pulmonary TB, adverse drug reactions, and unnecessary consumption of resources.

3.2.4. *Certainty of the evidence*

Judgments are more secure when one is certain about the evidence. Rating the certainty of the evidence for the effects of tests requires consideration of each element of the linked evidence used to inform judgments about their benefits and harms. EtD frameworks for tests include five

criteria for making judgments about the certainty of the evidence.

1. What is the certainty of the evidence of test accuracy?

A full GRADE evidence profile with detailed judgments about the certainty of the evidence about test accuracy should be used [16,30].

2. What is the certainty of the evidence for any critical or important test related direct benefits, adverse effects, or burden of the test?

Tests can bear a risk of direct harm and burden from conducting the test and assessing the certainty of the evidence for such effects can influence a decision about using a test. For example, using radiological contrast dye bears the risk of immediate allergic reactions and long-term adverse effects of radiation exposure. Use of CUS compared to contrast dye with intravenous pyelograms avoids this risk. Often this evidence comes from test accuracy studies, but it may come from other studies that evaluated the use of the test.

3. What is the certainty of the evidence of the natural course of the condition and the effects of management guided by the test results?

The certainty of prognostic evidence, such as for the natural course of a condition, can be assessed using GRADE [31]. Appendix 3 illustrates how the natural history of a disease could be accounted for in an EtD framework when a screening test results as false negative. This differs from the certainty one can place in the effects of treatments. The GRADE approach for evaluating the certainty of the evidence of effects for clinical interventions is now widely used by guideline developers [32] and a detailed description of this approach can be found elsewhere [33]. In the scenario described in Appendix 2, there is moderate certainty of the evidence for the health effects of heparin on DVT. Although only one small trial conducted almost 60 years ago evaluated the use of heparin and an oral anticoagulant in the treatment of PE, subsequent and not seriously indirect evidence confirms the reduction in the occurrence of DVT or evolution of DVT to PE in various populations at risk. Each body of evidence informing the pathway in a EtD focusing on tests will have a degree of certainty associated with it. The purpose of the EtD framework is to highlight, in a transparent way, this degree of certainty based on different bodies of evidence.

4. How certain is the link between test results and management decisions?

It is important to consider whether people with a given test result will be managed according to that result and how certain decision makers are about this or how certain the link is to evidence from prognostic studies. The scenario we presented earlier included the assumption that patients with suspected DVT who have a positive CUS result will receive heparin or other blood thinners. In this case, there is probably little doubt about this assumption, since the intervention is relatively simple to apply and few patients would be missed in health care systems that are equipped to offer testing. Thus, a panel might reasonably judge that there is high certainty about this link. In contrast, a WHO guideline panel evaluating screening for HIV thought it was less certain that a screened patient would receive appropriate services in a timely fashion and that those services would be available locally [34].

5. What is the overall certainty of the evidence about the effects of the test?

The overall rating of the certainty of the evidence about the effects of testing and subsequent management decisions on patient-important outcomes should be based on the certainty of the evidence for the weakest link in the chain of evidence used to estimate those effects. For example, a panel might decide that both the natural history of the disease and the effectiveness of the linked intervention were required for their estimates of the desirable and undesirable effects of using a test. In this case, despite high certainty of the evidence for test accuracy, the overall certainty of the evidence for their recommendation would be low if the evidence for the natural history of the disease or the effectiveness of the linked interventions was low.

3.2.5. Values

This criterion addresses how much people value the main outcomes and how that might affect decisions. For tests, this includes adverse effects and any burden associated with the test, as well as downstream outcomes of linked interventions. For example, some patients are highly adverse to blood sampling and intravenous lines and others do not mind as much. Patients who fear closed spaces will value not having an magnetic resonance imaging and might prefer a computed tomography (CT) scan in an open CT. Variability such as this is important for a recommendation when people who value the outcomes differently would make different choices.

3.2.6. Balance between the desirable and undesirable effects

Judgments about the balance between the desirable and undesirable effects of an intervention are critical for most recommendations and decisions. For tests, this judgment is informed by the results of either formal or informal modeling of the anticipated desirable and undesirable effects of linked interventions. The scenario in Box 1 introduces an example in which informal modeling informed a panel's judgment (Appendix 2).

3.2.7. Resource use

Judgments about resource use are the same as for other interventions. This includes judgments about how large the

resource requirements (costs) are, the certainty of the evidence of resource requirements and the cost-effectiveness of interventions. This includes consideration of downstream costs and can include issues such as opportunity cost if decisions are made from a health system perspective. For the scenario in Box 1, a published cost-effectiveness analysis from the United Kingdom of 23 noninvasive diagnostic testing strategies for DVT found that the most cost-effective strategy was to discharge patients with a low clinical probability of DVT and negative D-dimer without further testing, and to use ultrasound, with venography in selected cases, to diagnose the remaining patients [35]. In this case, a recommendation for or against using D-dimer to triage patients with suspected DVT, and other tests may be driven by resource use.

3.2.8. Equity, acceptability, and feasibility

For tests, assessments of equity, acceptability, and feasibility include consideration of both the test and linked interventions. For the scenario in Box 1, the panel did not find any evidence describing impacts on equity and judged that either diagnostic strategy would likely be acceptable to stakeholders. The panel did not identify important barriers to implementing the use of D-dimer, ultrasound and treatment, and judged that it would be feasible to implement use of D-dimer.

3.3. Drawing conclusions

Recommendations should follow from the judgments of the criteria used in the assessment. They follow from the evaluation of the criteria in the EtD, which of these criteria drive a recommendation can vary. In the scenario in Box 1, the balance of the desirable and undesirable consequences favored highly sensitive D-dimer testing followed by CUS if D-dimer positive and watchful waiting if D-dimer negative over D-dimer testing followed by whole leg ultrasound or venography if D-dimer positive or watchful waiting if D-dimer negative. The identified cost-effectiveness analysis addressing resource use also favored using D-dimer first, followed by CUS in patients who tested positive. Because there was low to moderate certainty of the evidence for test accuracy and some uncertainty about the effects on patient-important outcomes and costs (see Appendix 2), the panel made a conditional (weak) recommendation. This situation—moderate or high certainty in some elements, for example, test accuracy and treatment effects, but low or very low certainty in the evidence for all patient-important outcomes—is common, whenever estimates of the effects on patient-important outcomes are based on linking evidence and modeling.

This uncertainty, and the need for formal modeling, can be critical for screening tests. Appendix 3 shows the use of a formal model illustrating both the uncertainty about the size of effects, the value of explicit estimates, and a model that makes explicit the underlying evidence and assumptions.

4. Discussion

EtD frameworks for diagnostic and other tests share the requirement for transparency of other EtD frameworks and decision-making processes. They differ primarily in that they include additional criteria that address test accuracy and detailed assessments of the certainty of the evidence for the different types of evidence that are necessary. When there is no direct evidence of the effects of tests on important outcomes, modeling is needed to estimate these effects. Modeling can be formal (e.g., using evidence of effects and calculations in a decision analysis) or informal (e.g., making assumptions about effects with or without calculating impact on outcomes and presenting them in summary of findings tables or evidence profiles without formal models) [30]. The certainty that one can place in the evidence resulting from modeling is only as high as the weakest link between test accuracy and a patient or population important outcome.

Decisions about tests are complex with or without EtD frameworks. Our experience with the EtD frameworks suggests that they facilitate decision-making processes for guideline panels [14,22,23]. In particular, EtD frameworks for recommendations about tests lay out the sequential steps in assessing the evidence that needs to be linked, making transparent the different types of evidence and the certainty of this evidence. They highlight the importance of having an analytical framework describing the purpose and downstream effects of tests [25]. EtD frameworks are a result of teamwork of individuals or groups with different tasks. For example, in the guideline development context, the groups and tasks include those providing or conducting the evidence summaries for the questions a guideline panel has identified, the guideline panel that makes provides additional information and makes the judgments about the decision criteria and formulates the recommendation. These groups and tasks can be directly derived from the Guideline International Network—McMaster checklist on guideline development [36]. The EtDs, containing the description of the research evidence, can be either presented with preliminary judgments or with the expectation that the judgment are completed by those to whom they are presented, for example, a guideline panel. They are available in the GRADEpro Guideline Development Tool (www.gradepro.org), now as interactive EtDs with interactive SoF Tables. For example, in two recent World Health Organization Guidelines, the EtD frameworks were projected during guideline development meetings to panel members for their input, final judgment and conclusions before formulating a recommendation. Guideline developers may perceive the work required to prepare EtD frameworks as a weakness. However, this is not a limitation of the EtD framework itself, but a requirement for transparency in synthesizing evidence of test accuracy with other evidence, and in making judgments.

The goal for GRADEpro is to make evidence profiles and resulting guidelines with interactive EtDs available to others. The database of evidence profiles resulting from GRADE's DECIDE project (http://dbep.gradepro.org/) supports this effort and will facilitate use by others and adoption or adaptation of decisions from existing trustworthy guidelines.

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Supplementary data

Supplementary data related to this article can be found at http://dx.doi.org/10.1016/j.jclinepi.2016.01.032.

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